

# ECSP3000 Project Brief

*Designing and Implementing an Automated & Reproducible  
Testbed for Linux Rootkit Detectors, Aiming to Identify and  
Implement a Novel Rootkit Technique*

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## 1. Project Aims

Current research into the evaluation of rootkits and rootkit detectors rely on bespoke evaluation methods which cannot be reliably reproduced by third parties.

This tool aims to provide developers, researchers and system administrators with an automated and reproducible testbed for rootkits & rootkit detectors on the modern Linux kernel to aid future development and testing.

Furthermore the testbed aims to provide greater transparency on what rootkit techniques remain evasive to standard detection tools.

As an extension, I will use this tool to identify and evaluate current avenues in the modern (7.2) Linux Kernel for rootkit design, finishing with an evaluation using the aforementioned tool of a new rootkit PoC.

## 2. Project Objectives

This project will have the following aims which will be used to gauge success:

1. To have a testing framework which, upon specifying a set of rootkits & rootkit detectors, provides timely & accurate results which can then be reproduced exactly

upon rerun. z

2. The system will aim to treat rootkits and rootkit detectors as pluggable modules, allowing developers to easily test new rootkit detectors or test against new rootkits.
3. (Stretch) To research and develop a rootkit PoC for the modern 7.2 Linux Kernel, following with an evaluation using this testing framework.
4. Using a dataset of results produced by the testbed or, if successful, the aforementioned PoC, to evaluate the current status of rootkit detection on the modern Linux Kernel.

### 3. Limitations & Scope

In order to scope my project appropriately, here are a list of things that I currently plan on **not** doing.

1. My testbed and therefore my rootkit research will only be concerned with Kernel Space Rootkits. This excludes user-space, firmware or any other rootkit classes from my research.
2. Testing rootkits that implement anti-sandbox measures fall out of scope for this project due to the virtualised environment employed by the testing framework.